

Vidush Singhal

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📄 [vidush-singhal](#) | 🐙 [vidsinghal](#) | 🎓 [Google Scholar](#) |

707 N 5th St, Lafayette, IN, 47901

EDUCATION

- **Purdue University**, West Lafayette, IN, USA
Ph.D. in Electrical and Computer Engineering May 2021 – May 2027 (Expected)
- **Purdue University**, West Lafayette, IN, USA
M.S. in Electrical and Computer Engineering May 2021 – May 2023
- **Purdue University**, West Lafayette, IN, USA
B.Sc. in Computer Engineering Aug 2017 – May 2021

RESEARCH INTERESTS

I am a Ph.D. student in Electrical and Computer Engineering at Purdue University working on compiler techniques for improving the performance of programs over irregular recursive datatypes. My thesis is *Compiler Techniques for Optimizing the Layout of Irregular Recursive Datatypes*. My research interests include compilers, programming languages, program analysis, static analysis, data layout optimization, locality optimization, SIMD vectorization, automatic parallelization, and compilers for GPUs. I have experience building compiler infrastructure in Gibbon, LLVM, Clang, and MLIR/Torch-MLIR.

EXPERIENCE

- **Microsoft Research** [🌐] May 2025 – August 2025
Research Intern Redmond, WA, USA
 - Developed **C2Pulse**, an LLVM/Clang-based transpiler for verifying memory correctness of annotated C programs.
 - Implemented a **Clang-to-Clang** source transformer (~7K LOC of C++) that emits Pulse code for F* verification.
 - Built a C benchmark suite and validated generated verification conditions on representative annotated C programs.
 - Implemented and verified components of **DPE (Dice Protection Environment)** written in C through C2Pulse.
- **Lawrence Livermore National Laboratory** [🌐] May 2024 – August 2024
Computing Scholar Intern Livermore, CA, USA
 - Engineered an LLVM Attributor pass to **optimize byte layout within stack allocations** for improved spatial locality.
 - Implemented **LLVM IR** analyses and transformations to eliminate dead bytes and reorder frequently accessed bytes.
 - Optimized the **LLVM GPU sanitizer** by hoisting redundant checks out of loops, reducing XSBench runtime by **6%**.
 - Built an LLVM IR basic-block permutation pass to evaluate performance sensitivity to code layout.
- **Nod.ai Labs** [🌐] May 2022 – August 2022
Machine Learning Compiler Engineering Intern Santa Clara, CA, USA
 - Contributed to **Torch-MLIR**, a compiler stack for lowering programs from the **PyTorch ecosystem to MLIR**.
 - Implemented kernels and compiler support for multiple **PyTorch operators** in the Torch-MLIR pipeline.
 - Added support for the **Facebook DLRM** recommendation model in the PyTorch-to-MLIR compilation pipeline.
 - Investigated improvements to the **ILP solver** used by the Alpa scheduling framework.
- **Astrome Technologies Private Limited** [🌐] May 2019 – August 2019
Engineering Intern Bengaluru, Karnataka, India
 - Integrated an eMMC IP block with an AXI-Lite-to-Wishbone bridge using a SystemVerilog wrapper.
 - Verified the eMMC IP and bridge integration through Vivado behavioral simulations.
 - Built XSLT and Python tooling to render XML data as internal HTML reports.
 - Implemented convolutional encoding and Viterbi decoding algorithms in C for error correction and detection.
 - Achieved an **order-of-magnitude speedup** over the original Python implementation.
- **Purdue University** [🏠] May 2021 – Present
Graduate Research Assistant West Lafayette, IN, USA
 - **Gibbon Compiler**
 - * Optimized layouts of recursive datatypes to improve spatial locality.
 - * Measured speedups of **1.14x to 54x** over the best prior work.
 - * Implemented an array-of-structs to structure-of-arrays-style transformation for packed algebraic datatypes.

- * Structure-of-arrays transformation enables selective traversal, our benchmarks saw speedups of **1.1x to 12x**.
- * Developing optimizations enabled by the SoA representation such as loopification and vectorization.
- **Orchard Compiler**
 - * Developed a compiler framework to automatically fuse and parallelize tree traversals.
 - * Measured speedups ranging from **1x to 5x** over baseline tree traversals.
- **Copse Compiler**
 - * Developed a framework for vectorized evaluation of fully homomorphic encryption (FHE) decision trees.
- **Cornucopia**
 - * Developed a fuzzing-based framework to generate a corpus of optimized binaries from source programs using 11c optimization flags.
 - * Discovered bugs in LLVM compiler optimization passes.
 - * Exposed decompilation failures in angr, Ghidra, and related binary analysis tools on generated binaries.
 - * Embedded ground-truth metadata in generated binaries to evaluate binary analysis tools (BATs).
- **Senior Design Project, SoCET Team** [[🔗](#)] July 2020 – May 2021
West Lafayette, IN, USA
 - Senior Design Team Member; advised by Dr. Mark Johnson and Prof. Samuel Midkiff
 - Developed an LLVM-based compiler tool for proprietary RISC-V hardware.
 - Implemented multiple LLVM RISC-V backend Machine IR passes to identify sparse program regions.
 - Annotated sparse regions so hardware could detect them at runtime and perform runtime checks.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, T=THESIS

- [C.1] Raghav Malik, **Vidush Singhal**, Benjamin Gottfried, Milind Kulkarni "Vectorized Secure Evaluation of Decision Forests" in PLDI 2021 [[ACM DL](#)]
- [C.2] **Vidush Singhal**, Akul Abhilash Pillai, Charitha Saumya, Milind Kulkarni, Aravind Machiry "Cornucopia: A Framework for Feedback Guided Generation of Binaries" in ASE 2022 [[ACM DL](#)]
- [C.3] **Vidush Singhal**, Chaitanya Koparkar, Joseph Zullo, Artem Pelenitsyn, Michael Vollmer, Mike Rainey, Milind Kulkarni "Optimizing Layout of Recursive Datatypes with Marmoset" in ECOOP 2024 [[LIPICs](#)]
- [J.1] **Vidush Singhal**, Laith Sakka, Kirshanthan Sundararajah, Ryan Newton, Milind Kulkarni "Orchard: Heterogeneous Parallelism and Fine-grained Fusion for Complex Tree Traversals" in TACO 2024 [[ACM DL](#)]
- [C.4] Chaitanya S. Koparkar, **Vidush Singhal**, Aditya Gupta, Mike Rainey, Michael Vollmer, Artem Pelenitsyn, Sam Tobin-Hochstadt, Milind Kulkarni, Ryan R. Newton "Garbage Collection for Mostly Serialized Heaps" in ISMM 2024 [[ACM DL](#)]

COURSEWORK

Compilers	Compiler Code Generation and Optimization; Compilers for GPUs; Introduction to Compilers and Translation Engineering
Programming Languages	Reasoning About Programs; Programming Languages; Advanced C Programming; Object-Oriented Programming in C++
Algorithms & Theory	Computation Models and Methods; Applied Algorithms; Data Structures in C; Formal Languages, Computability, and Parallelization
Systems	Computer Architecture; Programming Parallel Machines; Microprocessors and Microcontrollers; Digital Design
Mathematics	Discrete Mathematics; Graph Theory; Category Theory; Random Variables
AI & Data Science	Introduction to Artificial Intelligence; Python for Data Science
Security & Quantum	Holistic Software Security; Introduction to Quantum Computing

SKILLS

Programming Languages	C; C++; Haskell; Python; Java; Bash; MATLAB; Assembly (ARM, x86); SystemVerilog
Compiler & Verification	LLVM; GCC; Coq; Dafny; Pulse; GDB
Parallel & HPC	CUDA; OpenMP; MPI; Pthreads; OpenCilk; Intel TBB; Intel Cilk Tools
Data & Machine Learning	PyTorch; TensorFlow; NumPy
Systems & Databases	Unix/Linux; Windows; PostgreSQL; AFL++; gem5
Web & Development Tools	HTML; CSS; Git; Visual Studio Code; Vim; CLion; Notepad++; Kate
Professional Skills	Communication; teamwork; inclusive collaboration

TEACHING EXPERIENCE

- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University Fall 2025
- **Teaching Assistant**, Prof. Milind Kulkarni, Intro to Data Science, Purdue University Spring 2023
- **Undergraduate Teaching Assistant**, Prof. Cheng Kok Koh, ECE 368, Purdue University Spring 2020
- **Student Grader for Signals and Systems**, Prof. Chih-Chun Wang, ECE 301, Purdue University Spring 2020

TALKS GIVEN

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- [MWPLS 2025](#), AoS To SoA Transformation of packed ADTs Fall 2025
 - [MWPLS 2024](#), Optimizing Layout of Recursive Datatypes with Marmoset Fall 2024
 - [ECOOP 2024](#), Optimizing Layout of Recursive Datatypes with Marmoset Fall 2024
 - [PurPL Seminar](#), Optimizing Layout of Recursive Datatypes with marmoset Spring 2024
 - [ASE 2022](#), Cornucopia: A Framework for Feedback Guided Generation of Binaries Fall 2022

HONORS AND AWARDS

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- Received a travel grant, LLVM developers meeting October 2024
 - Semester Honors, Purdue University Fall 2019 - Spring 2020
 - Deans's List and Semester Honors, Purdue University Fall 2017 - Spring 2018

LEADERSHIP EXPERIENCE

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- Seminar Co-Coordinator, Purdue Programming Languages and Systems Research Group (PurPL) Fall 2025 - present

REVIEW EXPERIENCE

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- Member of the Artifact Evaluation Committee (AEC), [ICFP 2026](#) Summer 2026
 - Member of the Program Committee (PC Member), [C3PO 2026](#) Spring 2026
 - Member of the Artifact Evaluation Committee (AEC), [PLDI 2026](#) Spring 2026
 - Member of the Artifact Evaluation Committee (AEC), [SPLASH 2026](#) Spring 2026
 - Additional Reviewer, [PPoPP 2026](#) Fall 2025
 - Member of the Reviews for the Journal of Open Source Software, [JOSS](#) Fall 2025
 - Member of the Artifact Evaluation Committee (AEC), [POPL 2026](#) Fall 2025
 - Member of the Artifact Evaluation Committee (AEC), [CGO 2026](#) Fall 2025
 - Member of the Artifact Evaluation Committee (AEC), [ICFP 2025](#) Summer 2025
 - Member of the Artifact Evaluation Committee (AEC), [PLDI 2025](#) Spring 2025
 - Collaborative Reviewer with Advisor (Milind Kulkarni), [CGO 2025](#) Spring 2025
 - Member of the Artifact Evaluation Committee (AEC), [CGO 2025](#) Fall 2024
 - Member of the Artifact Evaluation Committee (AEC), [PPoPP 2024](#) Spring 2024
 - Member of the Artifact Evaluation Committee (AEC), [PPoPP 2023](#) Spring 2023

VOLUNTEER EXPERIENCE

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- Talk moderator, LLVM developers meeting 2024 Fall 2024
 - Student Volunteer, ECOOP/ISSTA 2024 Spring 2024
 - Student Volunteer, SPLASH 2021 Fall 2021

PROFESSIONAL MEMBERSHIPS

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- Eta Kappa Nu, Purdue Electrical and Computer Engineering Honors Society Fall 2018 - Present

MENTORING

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- Logan Anderson, Phd @ Purdue Fall 2025
 - Aniruddh Srivastava, Undergrad @ Purdue Fall 2025
 - Peter A Kaya Gretchikha, Undergrad @ Purdue Fall 2024 - Fall 2025
 - Mikah Kainen, Undergrad @ Purdue Fall 2023
 - Qingyuan (Peter) Li, Undergrad @ Purdue, SURF summer research mentor Spring 2023 - Fall 2023